

Mathematical Explanation of Data Re-Uploading in Variational Quantum Circuits

1 Problem Setting

In the `sin_encoded` experiment, the target function is

$$y = \sin(2\pi x) + \varepsilon, \quad x \in [0, 1], \quad \varepsilon \sim \mathcal{N}(0, \sigma^2). \quad (1)$$

The model is a variational quantum circuit (VQC) whose prediction is an expectation value of a measured observable,

$$f_{\boldsymbol{\theta}}(x) = \langle 0|U^\dagger(x, \boldsymbol{\theta})OU(x, \boldsymbol{\theta})|0\rangle, \quad (2)$$

where O is a Pauli observable and $\boldsymbol{\theta}$ are trainable parameters. Since Pauli expectation values lie in $[-1, 1]$, this output range is well matched to a sine-regression target.

2 Angle Encoding

A common way to load a classical scalar x into a quantum circuit is angle encoding. For one qubit, the feature map can be written as

$$S(x) = R_Y(x), \quad R_Y(x) = \exp\left(-i\frac{x}{2}Y\right). \quad (3)$$

After this rotation, the quantum state contains trigonometric dependence on the input:

$$R_Y(x)|0\rangle = \cos\left(\frac{x}{2}\right)|0\rangle + \sin\left(\frac{x}{2}\right)|1\rangle. \quad (4)$$

Therefore, any observable expectation built from this encoded state is a trigonometric function of x . However, if the input is inserted only once and at one frequency scale, the accessible Fourier spectrum is limited [2].

3 Data Re-Uploading

Data re-uploading increases expressivity by inserting the same classical input multiple times into the circuit, usually interleaved with trainable blocks [1]. A generic data re-uploading circuit has the form

$$U(x, \boldsymbol{\theta}) = W_L(\boldsymbol{\theta}_L)S_L(x) \cdots W_2(\boldsymbol{\theta}_2)S_2(x)W_1(\boldsymbol{\theta}_1)S_1(x), \quad (5)$$

where each $S_\ell(x)$ is a data-dependent encoding block and each $W_\ell(\boldsymbol{\theta}_\ell)$ is a trainable unitary. Repeatedly encoding the same input lets the trainable circuit combine several nonlinear trigonometric components.

In the `sin_encoded` notebook, the re-uploading is implemented across four qubits by first scaling the scalar input and then assigning one scaled copy to each qubit:

$$\tilde{x} = \text{MinMaxScale}(x) \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right], \quad (6)$$

$$\mathbf{z}(x) = (\tilde{x}, 2\tilde{x}, 3\tilde{x}, 4\tilde{x}). \quad (7)$$

The feature map is then

$$S(x) = \bigotimes_{k=1}^4 R_Y(k\tilde{x}). \quad (8)$$

Equivalently, qubit k receives the rotation angle $k\tilde{x}$. This creates several Fourier components of the same original scalar input.

4 Fourier View

The reason data re-uploading helps is that expectation values of parametrized quantum circuits with angle encodings can be represented as finite Fourier series in the input [2]. Abstractly,

$$f_{\boldsymbol{\theta}}(x) = \sum_{\omega \in \Omega} c_{\omega}(\boldsymbol{\theta}) e^{i\omega x}, \quad (9)$$

or, for real-valued outputs,

$$f_{\boldsymbol{\theta}}(x) = a_0(\boldsymbol{\theta}) + \sum_{\omega \in \Omega_+} [a_{\omega}(\boldsymbol{\theta}) \cos(\omega x) + b_{\omega}(\boldsymbol{\theta}) \sin(\omega x)]. \quad (10)$$

The set Ω of available frequencies is determined by how the input enters the circuit. With only a single encoding $R_Y(\tilde{x})$, the model can express only a small Fourier family. With the four re-uploaded angles $\tilde{x}, 2\tilde{x}, 3\tilde{x}, 4\tilde{x}$, the model has access to a richer frequency set. The trainable ansatz and entangling gates then learn the coefficients $a_{\omega}(\boldsymbol{\theta})$ and $b_{\omega}(\boldsymbol{\theta})$.

This is the quantum analogue of replacing a scalar feature x with classical Fourier features such as

$$\phi(x) = (\sin x, \cos x, \sin 2x, \cos 2x, \sin 3x, \cos 3x, \sin 4x, \cos 4x). \quad (11)$$

The circuit does not explicitly construct this classical vector. Instead, the feature map and subsequent quantum operations implicitly generate and combine these trigonometric components.

5 Why It Helps for Sine Regression

The target function $\sin(2\pi x)$ is itself a Fourier mode. A circuit that sees the input only through one weak angle encoding may not have enough spectral bandwidth to approximate the target accurately after scaling and noise are included. Data re-uploading expands the available basis functions, so the model can fit the dominant sine mode and use additional components to absorb finite-sample and noisy-data effects.

For the four-qubit encoding used in `sin_encoded`, the model input is not four independent features. It is one scalar x represented at four different frequency scales. The improvement therefore comes from a richer encoding of the same information, not from adding extra data.

6 Conclusion

Data re-uploading makes the VQC more expressive by allowing the measured expectation value to contain a richer Fourier expansion of the input. In the `sin_encoded` experiment, the transformation

$$x \mapsto (\tilde{x}, 2\tilde{x}, 3\tilde{x}, 4\tilde{x})$$

is mathematically appropriate for a sine-regression task because it gives the quantum model access to multiple trigonometric frequencies of the same scalar input.

References

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